

Sparse Counterfactual Attribution of Crop-Yield Anomalies to Extreme-Weather Drivers

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Abstract

Machine-learning studies of crop yield usually optimize average predictive accuracy, but agricultural risk management also needs event-level explanations: why did a particular crop-state-year fall below its expected yield? We present Sparse Counterfactual Anomaly Attribution (SCAA), a model-based framework that detects detrended low-yield anomalies and explains them using sparse, physically grouped extreme-weather edits. Daily NASA POWER weather and USDA yield records are aggregated for four crops, twelve U.S. states, and 1990–2025. A forward-time ExtraTrees yield model attains $R^2 = 0.809$ for 2016–2021 and $R^2 = 0.808$ for 2019–2025. After crop-state detrending, 214 of 1,257 observations are flagged as low-yield anomalies. The main submission protocol is retrospective leave-one-event-year-out grouped SCAA: for every anomaly year, the residual model is fitted after excluding all rows from that year. Under this conservative protocol, recovery is intentionally lower than in-sample recovery, reflecting the difficulty of residual-scale anomaly attribution when the event year is excluded from model fitting. Event-year consistency is reported as a limited sanity check rather than primary validation because broad heat/drought labels are also matched by simple null baselines. The contribution is a reproducible workflow for transparent retrospective model-based weather-driver attribution on a detrended yield-residual scale, not a new yield prediction model or high-skill residual forecasting system.

Keywords: crop yield; extreme weather; counterfactual explanation; yield anomaly; interpretable machine learning; drought; heat stress.

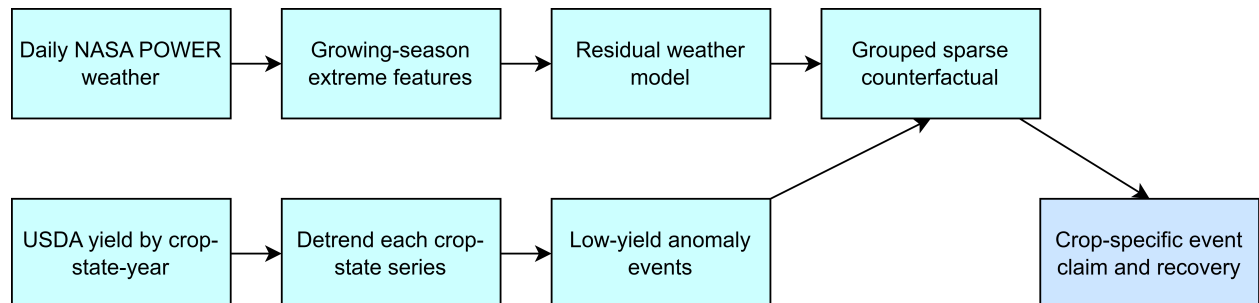
1 Introduction

Extreme weather can produce large agricultural losses, with drought, heat, excess water, and cold events affecting crop production at regional and global scales [7, 32, 34]. Standard crop-yield machine-learning workflows demonstrate that weather and climate variables can support accurate yield forecasting [5, 6, 11, 20], but prediction alone does not answer the event-level question needed for risk interpretation: which weather driver does the model identify as the strongest explanation for a specific low-yield crop-state-year?

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This paper reframes yield interpretation as anomaly attribution. We first separate long-run crop-state yield trend from short-run variation, following detrended yield residual analysis [8, 10, 22]. We then detect low-yield anomalies and ask a counterfactual question on the raw residual scale: what sparse, feasible change to a weather-driver group would move the predicted detrended residual toward a normal season? Unlike feature-importance methods such as SHAP or LIME [9, 23], this yields an event-level recovery quantity in t ha^{-1} and a crop-specific claim.

The contribution is not a new yield predictor. The contribution is a reproducible workflow that combines detrended yield anomalies, retrospective leave-one-event-year-out residual modelling, physically grouped extreme-weather drivers, and sparse counterfactual recovery. Figure 1 summarizes the pipeline.



SCAA explains detrended low-yield events with sparse, feasible, physically grouped weather changes.

Figure 1: SCAA workflow. Daily weather and yield records are converted into crop-state-year features, detrended anomalies, retrospective leave-one-event-year-out residual models, and grouped counterfactual recovery estimates.

2 Related Work

Crop-yield forecasting has been widely studied using statistical models, random forests, neural networks, and process-informed machine-learning approaches, showing that weather and climate variables contain substantial predictive signal for agricultural production [5, 11, 20, 25]. A related literature connects yield anomalies to extreme-weather stressors such as heatwaves, drought, excess rainfall, frost, and compound hot-dry events, with effects often varying by crop type, latitude, and growing stage [4, 24, 27, 32, 34]. However, most forecasting studies emphasize average predictive accuracy or aggregate feature importance rather than explaining why a specific crop-state-year became a low-yield anomaly. Counterfactual explanations provide a useful event-level form by asking what minimal or feasible feature changes would alter a model prediction [12, 21, 29, 31, 33], while climate-event attribution highlights the need for careful event definition and caution about causal claims [3, 19, 30]. This paper connects these strands by using detrended yield anomalies, physically grouped weather drivers, and sparse counterfactual recovery estimates to produce model-internal attribution claims for individual crop-state-year events.

3 Data

The empirical unit is a crop-state-year. The processed frame contains 1,257 observations for Barley, Canola, Oats, and Wheat across twelve U.S. states from 1990–2025. Yields are in t ha^{-1} and are

derived from USDA NASS yield records [28]. Weather features are derived from NASA POWER daily records [13] and include rainfall totals, dry spells, heavy-rain events, heat days, heatwaves, growing-degree days, frost/cold metrics, and radiation summaries.

The weather extraction uses one representative latitude-longitude coordinate per state. Thus, all crops in a state share the same daily weather series before crop-specific growing-season windows are applied. The design is transparent and reproducible, but it should be interpreted as representative-coordinate state weather rather than a crop-area-weighted state average or a within-state grid average. Growing season windows are crop- and state-aware, with spring and winter windows represented in the processed frame.

Table 1: Dataset summary by crop.

crop	observations	states	years	windows	anomalies
Barley	283	10	1990-2025	spring	52
Canola	133	6	1991-2025	spring	21
Oats	416	12	1990-2025	spring	71
Wheat	425	12	1990-2025	spring, winter	70

Table 1 shows that the processed panel is moderately unbalanced across crops. Oats and Wheat provide the largest number of crop-state-year observations, while Canola has fewer observations and fewer states. This motivates reporting event-level claims with crop and state identifiers rather than aggregating all crops into a single global attribution statement.

Table 2: Extreme-weather driver groups used by grouped SCAA.

driver_group	description
heat	High-temperature and heatwave exposure
drought	Rainfall deficit, dry days, and dry-spell persistence
frost_cold	Cold and frost exposure
excess_rain	Wetness and heavy-rainfall intensity
radiation	Seasonal solar-radiation anomaly

The driver groups in Table 2 are used to avoid over-interpreting individual weather variables. Each counterfactual edit changes a physically related group of features, such as rainfall deficit and dry-spell persistence for drought, rather than selecting isolated variables independently. This grouping makes the diagnostic explanation more interpretable and reduces the risk of feature-level cherry-picking.

4 Method

4.1 Yield Skill Baseline

We retain a forward-time yield model to verify that the weather and crop-state features contain predictive signal. Numeric variables are median-imputed and standardized; categorical crop and

state identifiers are one-hot encoded. An ExtraTrees regressor is trained under two forward-time splits: train through 2015 and test 2016–2021; then train through 2018 and test 2019–2025.

4.2 Detrending and Anomaly Screening

For each crop-state series, a linear trend is fit to yield:

$$y_{c,s,t} = \alpha_{c,s} + \beta_{c,s}t + \epsilon_{c,s,t}. \quad (1)$$

The raw detrended residual is

$$r_i = y_i - y_i^{trend}, \quad (2)$$

with unit t ha^{-1} . The standardized residual z_i is used only to screen low-yield anomalies. An observation is flagged when $z_i < -1.0$. We use the term “low-yield anomaly” rather than “extreme anomaly” because this threshold is meant to capture a broad set of below-trend events for attribution.

4.3 Retrospective Leave-One-Event-Year-Out Grouped SCAA

The main submission method is retrospective leave-one-event-year-out grouped SCAA. For each anomaly year t^* , we train a residual model f_{-t^*} on all crop-state-year rows except those from t^* . This prevents the model from directly learning the same event year being explained. It is not a fully prospective forecast, however, because observations after t^* may remain in the training set. The target is the raw detrended yield residual `trend_residual_t_ha`, not the standardized residual. The fitted model maps weather features, crop, and state to predicted residual $\hat{r}_i$ on the same t ha^{-1} scale.

This protocol is designed for retrospective attribution rather than operational forecasting. Its purpose is to prevent the residual model from directly fitting the same calendar-year shock that is later explained. Thus, if a drought year affects many crop-state observations simultaneously, all rows from that year are removed before the counterfactual recovery for that year is computed. This makes the submitted recovery estimates more conservative than in-sample explanations.

For an anomaly i , grouped SCAA evaluates driver groups one at a time. For a group g , all weather features in that group are moved toward their observed median for the same region-window support, giving a diagnostic edited feature vector $\mathbf{x}_i^{cf(g)}$. The observed detrended yield shortfall is

$$\Delta_i = \max(0, y_i^{trend} - y_i). \quad (3)$$

The recovered yield gap for the best group edit is

$$\text{recovered_gap}_i = \min \left(\max(0, \hat{r}_i^{cf} - \hat{r}_i), \Delta_i \right), \quad (4)$$

and the recoverable fraction is

$$\phi_i = \begin{cases} \text{recovered_gap}_i / \Delta_i, & \Delta_i > 0, \\ 0, & \Delta_i = 0. \end{cases} \quad (5)$$

A crop-state-year is labelled a high-recovery diagnostic case when $\phi_i \geq 0.5$. The anomaly flag uses the standardized residual z_i , whereas all SCAA recovery quantities use raw detrended residual and yield-gap units of t ha^{-1} .

The grouped edit is a diagnostic sparse edit designed to isolate one physical driver group. It is not assumed to be a complete season that occurred historically. For that reason, we also report

observed-analog counterfactuals: real non-anomalous seasons from the same crop-state, or the same region-window fallback, used as a plausibility check for weather-vector replacement.

In summary, grouped SCAA proceeds in five steps: (1) detrend each crop-state yield series and screen low-yield anomalies using standardized residuals; (2) for each anomaly year, train a residual model after excluding all rows from that year; (3) for each anomaly and each driver group, replace the group features by feasible region-window medians while holding other features fixed; (4) compute the bounded improvement in predicted raw residual relative to the observed yield gap; and (5) assign the driver group with the largest recoverable fraction as the dominant diagnostic explanation.

4.4 Residual-Model Validation

Because grouped SCAA operates on the raw detrended residual scale, we separately evaluate the residual model used for attribution. We report R^2 , RMSE, and MAE for `trend_residual_t_ha` under a forward-time residual split and under the same retrospective leave-one-event-year-out protocol used by grouped SCAA. These metrics do not replace the forward-time yield model; they verify how much residual-scale signal is available before diagnostic counterfactual recovery is interpreted.

4.5 Robustness and Extensions

We include four robustness or extension layers. First, in-sample grouped SCAA is kept as an exploratory comparison to show the effect of removing year holdout. Second, observed-analog counterfactuals test whether recovery is possible using historically observed normal seasons. Third, anomaly-threshold and detrending sensitivity tables check whether the anomaly set is an artifact of a single screening choice. Fourth, crop-specific vulnerability profiles stress-test each crop only on observed crop-state-year combinations. An early-warning extension trains anomaly classifiers from early and early-plus-mid season weather features, with calibration-derived uncertainty bands [1, 2, 26].

5 Results

5.1 Yield Skill and Anomalies

The yield model reproduces strong forward-time predictive skill (Table 3). The 2016–2021 split attains $R^2 = 0.809$ and RMSE 0.525 t ha^{-1} ; the 2019–2025 split attains $R^2 = 0.808$ and RMSE 0.536 t ha^{-1} . After detrending, 214 of 1,257 observations are flagged as low-yield anomalies. The anomaly timeline highlights 2012, 2021, and 2022 as event years used for the consistency check (Figure 2).

Table 3: Forward-time yield performance and anomaly count.

evaluation	test_years	n_test	r2	rmse_t_ha
test_2016_2021	2016-2021	209	0.809	0.525
test_2019_2025	2019-2025	236	0.808	0.536
window_spring_2016_2021	2016-2021	164	0.842	0.475
window_winter_2016_2021	2016-2021	45	0.543	0.677
low_yield_anomalies	1990-2025	1257		214 anomalies

The forward-time yield results are used only as a signal check. They show that the weather and crop-state features contain predictive information for yield levels, but they are not used as evidence that residual anomaly magnitudes can be forecast with high accuracy. The attribution task is therefore evaluated separately on the detrended residual scale.

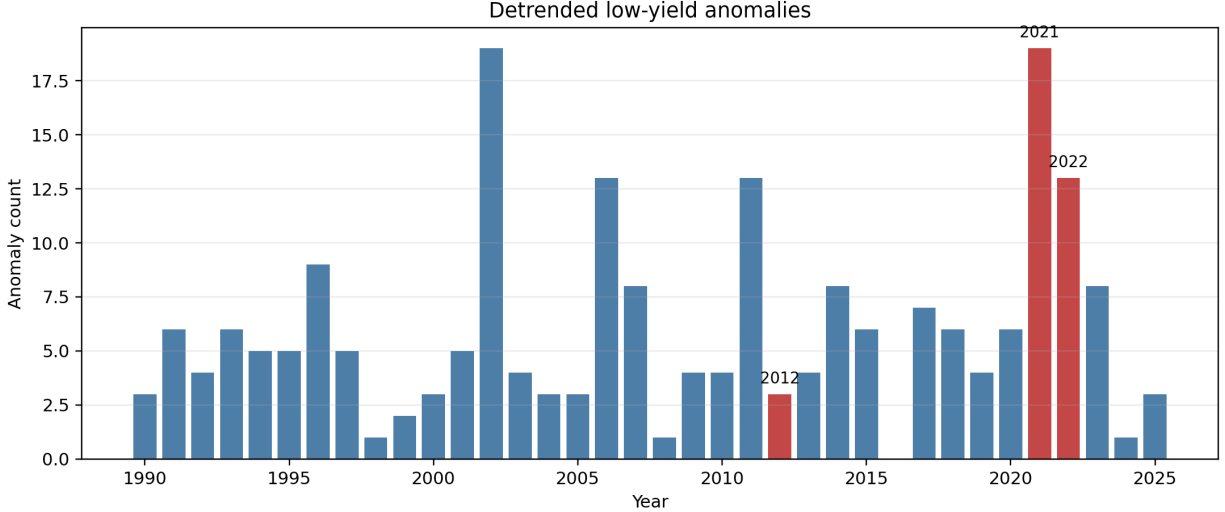


Figure 2: Low-yield anomaly counts by year, with 2012, 2021, and 2022 highlighted.

5.2 Residual-Model Validation

Table 4 reports the predictive skill of the residual model used by grouped SCAA. The forward-time residual split has $R^2 = 0.093$, RMSE 0.431 t ha^{-1} , and MAE 0.326 t ha^{-1} . Under retrospective leave-one-event-year-out evaluation across all held-out rows, $R^2 = 0.085$, RMSE is 0.369 t ha^{-1} , and MAE is 0.284 t ha^{-1} . On the anomaly rows alone, R^2 is negative ($R^2 = -5.152$), indicating that anomaly-level residual magnitudes are difficult to predict directly. We therefore interpret SCAA as a diagnostic recovery framework with bounded counterfactual edits, not as a high-skill residual forecasting model.

The residual-model validation results define the scope of the method. Residual-scale prediction is modest across all held-out rows and weak on anomaly rows, which means that SCAA should be interpreted as a conservative diagnostic recovery tool rather than a high-skill residual forecasting system. This is why the manuscript reports bounded recovery quantities and avoids causal or operational forecasting claims.

Table 4: Residual-model validation for the attribution target.

protocol	held_out_unit	n_eval	r2	rmse_t_ha	mae_t_ha
Forward-time residual	2016-2025	343	0.093	0.431	0.326
Leave-one-event-year-out residual	All held-out rows	1222	0.085	0.369	0.284
Leave-one-event-year-out residual	Anomaly rows	214	-5.152	0.562	0.496

5.3 Method Comparison

Table 5 compares methods by raw metrics only. The retrospective leave-one-event-year-out method is more conservative than the in-sample grouped method: median recoverable fraction decreases from 0.589 to 0.116, and the high-recovery rate decreases from 73.4% to 5.6% (12 of 214 anomalies). This reduction is expected because each event year is excluded before attribution. We therefore use leave-one-event-year-out grouped SCAA for the main submission claims, while the in-sample grouped result is treated as exploratory. Observed-analog counterfactuals show larger recovery because they replace the full weather vector with a real normal season; they are used as a plausibility comparison rather than the main sparse explanation.

Table 5: Comparison of attribution methods using recovery, high-recovery rate, and event-year agreement.

method	median_recoverable_fraction	high_recovery_rate	event_expected_match_rate
Raw-yield SCAA	0.379	0.262	0.829
Residual-target SCAA	0.482	0.472	0.629
Grouped-driver SCAA	0.589	0.734	0.886
Leave-one-event-year-out grouped SCAA	0.116	0.056	0.914
Observed-analog counterfactual	0.853	0.972	0.971

5.4 Leave-One-Event-Year-Out Grouped Attribution

Retrospective leave-one-event-year-out grouped SCAA assigns a recovery fraction and dominant driver group to each low-yield anomaly (Figure 3). The median recoverable fraction is 0.116 and the mean is 0.157. Drought/dry-spell stress is the most frequent driver group, followed by heat exposure, excess rainfall or wetness, frost/cold stress, and radiation. The highest-recovery crop-state-year diagnostic cases are shown in Table 6. For example, South Dakota Oats in 2006 is attributed to drought/dry-spell stress, with a modelled recovery of 0.273 t ha⁻¹, or 71.7% of the observed detrended yield shortfall.

Table 6: Highest-recovery retrospective leave-one-event-year-out grouped-SCAA crop-state-year diagnostic cases.

crop	state	year	driver_group	dominant_feature	yield_gap_t_ha	recovered_gap_t_ha	recoverable_fraction
Oats	South Dakota	2006	drought	rain_mean	0.381	0.273	0.717
Barley	Washington	2015	drought	hot_dry_days_30_1mm	0.862	0.598	0.693
Wheat	Washington	1992	drought	max_dry_spell_1mm	0.579	0.39	0.672
Wheat	Texas	2009	drought	dry_spell_events_14d	0.41	0.273	0.666
Canola	North Dakota	2012	drought	hot_dry_days_30_1mm	0.285	0.183	0.64

These rows should be read as diagnostic model-based cases rather than causal event attributions. A high recoverable fraction means that the fitted residual model assigns substantial bounded recovery to one physical driver-group edit, not that the driver is the sole real-world explanation for the yield shortfall.

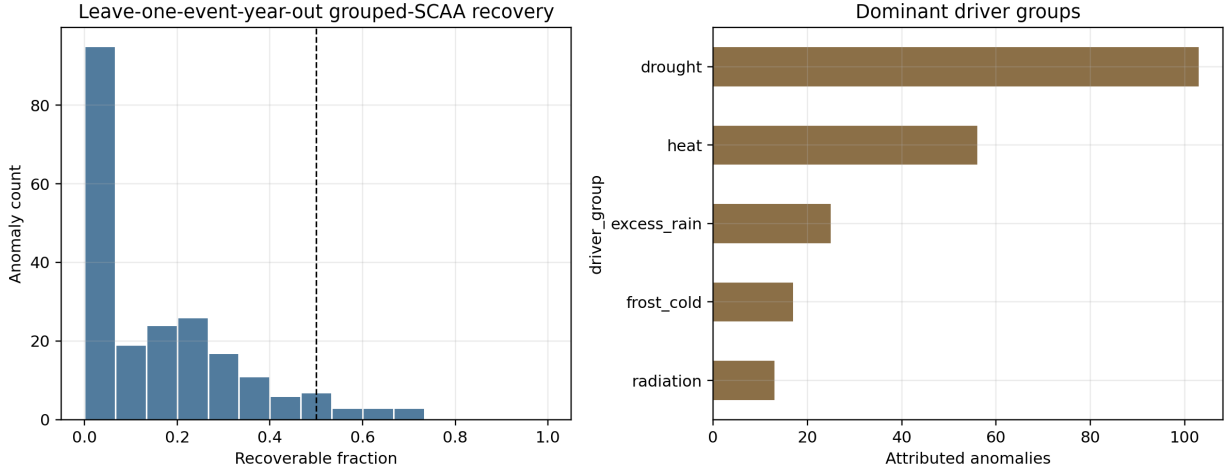


Figure 3: Retrospective leave-one-event-year-out grouped-SCAA recoverable fraction and dominant driver-group frequency.

Figure 3 shows that most anomaly events have limited recoverable fractions under the conservative protocol, while a smaller subset achieves substantial recovery. This pattern is useful because it prevents the method from forcing a weather explanation for every low-yield anomaly. The driver-frequency panel indicates that drought/dry-spell stress is the most common dominant group, followed by heat exposure and excess rainfall or wetness.

5.5 Event-Year Consistency Check

We pre-specified broad expected stress groups for 2012, 2021, and 2022 using external NOAA/NCEI, National Weather Service, and Drought.gov/NIDIS evidence [14–18]. This check asks whether the SCAA dominant groups are consistent with those external stress categories; it does not validate formal event drivers. Retrospective leave-one-event-year-out grouped SCAA matches the pre-specified expected groups in 91.4% of event-year rows: 100% in 2012, 100% in 2021, and 76.9% in 2022.

Because the event-year labels are intentionally broad, simple null baselines can also achieve high agreement. We therefore treat event-year consistency only as a sanity check that the diagnostic edits do not contradict externally documented stress categories, rather than as primary validation or evidence of superiority over trivial event labels. The null-baseline table, detailed event-evidence table, and event-year consistency panel are provided in the Supplementary Material to keep the main manuscript focused on the core diagnostic result.

5.6 Crop-Specific Vulnerability Profiles

The vulnerability profile answers which weather driver most reduces which crop in the fitted residual model, evaluated only on observed crop-state-year combinations. Table 7 reports positive yield penalties under adverse observed extremes, so larger values indicate stronger modelled vulnerability. Barley is most sensitive to heat exposure, with a median penalty of 0.134 t ha^{-1} ; Wheat is most sensitive to drought/dry-spell stress, with a median penalty of 0.103 t ha^{-1} ; and Canola is most sensitive to heat exposure, with a median penalty of 0.086 t ha^{-1} .

Table 7: Top crop-specific yield penalties under adverse observed weather extremes.

crop	driver_group	median_yield_penalty_t_ha	states_largest_penalty
Barley	heat	0.134	Oklahoma (0.253); South Dakota (0.222); Kansas (0.191)
Wheat	drought	0.103	Washington (0.378); Colorado (0.327); Oklahoma (0.282)
Canola	heat	0.086	Kansas (0.245); Minnesota (0.167); Oklahoma (0.142)

Early-warning metrics and the early-versus-mid-season comparison figure are provided in the Supplementary Material. The extension is exploratory because anomaly labels and detrending are defined retrospectively in this version.

6 Discussion

The revised results support a conservative paper framing. The retrospective leave-one-event-year-out SCAA protocol produces lower recovery than the in-sample grouped method, but it better addresses external-by-year attribution concerns because the event year is absent from the residual-model fit. Its strongest contribution is therefore not high recovery for every anomaly; it is a transparent event-level protocol that reports crop, state, year, driver group, dominant feature, recovered $t \text{ ha}^{-1}$, and the residual-scale recovery formula.

The weak anomaly-row residual R^2 (-5.152) defines the scope of interpretation. SCAA should not be used to forecast anomaly magnitudes prospectively. Its narrower role is to ask, for a screened below-trend event and a fitted residual model, which physically grouped weather edit produces the largest bounded movement toward the trend line. For this reason, each event is reported with crop, state, year, driver group, dominant feature, recovered $t \text{ ha}^{-1}$, and recoverable fraction, rather than as an unconditional real-world explanation.

The threshold sensitivity analysis shows that the anomaly count changes from 214 at $z < -1.0$ to 85 at $z < -1.5$ and 30 at $z < -2.0$, while the leave-one-event-year-out median recoverable fraction remains close to 0.12 across thresholds. Detrending robustness is moderate: quadratic detrending has Jaccard overlap 0.774 with the linear anomaly set, while a centered seven-year rolling baseline has overlap 0.550. We keep linear detrending because it is transparent and low-parameter, but report these alternatives to make the screening choice explicit.

There are several limitations. The counterfactuals are model-internal predictive associations and should not be read as formal climate-event attribution. State-level representative-coordinate weather hides within-state variation in weather, irrigation, soil, cultivar, and management. The detrending step is also estimated over the full 1990–2025 retrospective crop-state series; an operational early-warning deployment would require rolling or past-only trend estimation before anomaly screening. The grouped edit changes several related features toward medians and may not represent a physically complete weather season; observed-analog counterfactuals partly address this by using historical seasons that actually occurred. Finally, the early-warning extension uses realized early-season weather, not operational seasonal forecasts.

7 Conclusion

SCAA detects detrended low-yield anomalies and explains them through sparse, physically grouped weather counterfactuals on a raw residual scale. On U.S. Barley, Canola, Oats, and Wheat across 1990–2025, the retrospective leave-one-event-year-out grouped-SCAA protocol produces conservative crop-state-year claims such as drought/dry-spell stress in South Dakota Oats 2006, with explicit recovered $t \text{ ha}^{-1}$ and recoverable fractions. The result is a reproducible bridge between crop-yield

prediction, interpretable machine learning, and cautious diagnostic event-level weather-driver analysis.

Additional driver-feature, robustness, event-consistency, and early-warning materials are provided in the Supplementary Material.

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